

Johnny Adrián García Betancourt

M.Sc. Researcher in Electrical Engineering | Geometric Mechanics • Nonlinear Dynamics • Physics-Guided ML

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Research Profile

Mechatronics engineer and M.Sc. researcher studying the geometric, dynamical, and energetic structure of nonlinear physical systems and how it can guide machine-learning models. My work combines differential geometry, port-Hamiltonian modelling, passivity, differential flatness, and deep learning for physics-guided representation learning and fault diagnosis. I aim to extend this structural viewpoint toward Scientific Machine Learning and Scientific AI by incorporating physical laws, geometry, symmetries, and energetic structure into learning models.

Education

M.Sc. Student in Electrical Engineering – Control & Robotics Track 2025–Present
King Fahd University of Petroleum & Minerals (KFUPM), Saudi Arabia GPA: 3.938/4.00

Thesis: Bridging Physical Safety and Active Tolerance in Robotic Manipulators: A Hierarchical Energy-Aware Architecture.

Selected Coursework: Classical Mechanics (PHYS 507, A+), Advanced Methods for Theoretical Physics (PHYS 571, A+).

Graduate Coursework in Automation and Control (Master's Program) 2022–2024

Escuela Superior Politécnica del Litoral (ESPOL), Ecuador

Coursework record: 10 of 11 completed courses graded A+ | Average: 9.68/10

Completed advanced graduate coursework before transferring to KFUPM prior to thesis completion.

Selected Coursework: Machine Learning-Based Control (A+), Industrial Robotics & Computer Vision (A+).

B.Sc. in Mechatronics Engineering (Engineer in Mechatronics) 2017–2022

Escuela Superior Politécnica del Litoral (ESPOL), Ecuador

Thesis Grade: 9.20/10

Thesis: Mechanical and Electrical Design, and Automation of an Industrial Liquid Filling Machine.

Research Experience

Spectral-Hamiltonian Algebraic Power Evaluation (SHAPE Framework) Research Project | KFUPM

Conceived and formulated a physics-guided fault-diagnosis framework combining port-Hamiltonian power structure with differential flatness to construct observer-free residuals. Led the physical simulation and data-generation pipeline, co-designed the multi-joint CNN, and worked directly on its Python/PyTorch implementation using temporal, CWT, and spectral representations.

Hierarchical Energy-Aware Architecture for Safe Robotic Interaction M.Sc. Thesis | KFUPM

Developing a fault-tolerant control architecture for redundant manipulators that combines port-Hamiltonian modelling and energy tanks with geometric trajectory replanning. The work investigates energy freezing and regulated tank exchange while adapting motion on the configuration manifold as actuator failures progressively restrict the robot's reachable configurations.

Active Fault-Tolerant Flatness-Based Model Predictive Control Research Manuscript | Revised for Resubmission

Developed an active fault-tolerant MPC formulation that maps actuator degradation into dynamically consistent constraints in flat-output space. The method adapts feasible trajectory tracking as available control authority decreases, using differential flatness to translate physical actuator limits into algebraic constraints suitable for predictive optimization.

Geometric Modelling & Invariant State Estimation of an Underwater Glider Coursework & Research Projects | KFUPM

Modelled an underactuated underwater glider that achieves low-energy locomotion without conventional propulsive actuation, using internal mass displacement and variable buoyancy to modify its centre of gravity and effective density. Studied its nonlinear geometric dynamics and invariant state estimation in MATLAB/Simulink.

Self-Diagnosing Process Automation & Predictive Maintenance Graduate Research Project | ESPOL

Developed an automated liquid-processing system integrating recipe-based mixing, pumps, motors, pneumatic actuation, PLC/HMI control, and industrial communication through IO-Link. Integrated multi-parameter condition-monitoring sensors and explored MATLAB-based predictive-maintenance and machine-learning workflows for automated fault diagnosis and equipment health assessment.

Design, Modelling & Automation of an Industrial Liquid-Filling Machine B.Sc. Thesis | ESPOL

Designed and modelled an automated filling machine for viscous liquid products, integrating pneumatic actuation, fluid-flow calculations, electrical sizing, PLC/HMI control, and mechanical design. Performed coupled pneumatic/fluid calculations for dosing accuracy and cycle timing, and used CAD/CAE and finite-element analysis to dimension the liquid hopper and structural components.

Research Manuscripts & Working Papers

[1] **J. Garcia Betancourt** and N. Salguero España, “Active Fault-Tolerant Flatness-Based MPC for Robotic Manipulators under Parametric Mismatch and Actuator Degradation,” *Research Manuscript*, revised for resubmission, 2026.

[2] **J. Garcia Betancourt**, N. Salguero España, N. Iqbal, B. Brahmi, and R. Rashad, “Energy-Based Fault Detection and Isolation Framework for Flat Port-Hamiltonian Mechanical Systems,” *Research Manuscript*, in final preparation, 2026.

[3] U. I. Castilla Batun and **J. A. Garcia Betancourt**, “Techno-Economic Feasibility of Wind-Powered Hydrogen Production in Saudi Arabia: A Multi-Regional Assessment,” *Conference Manuscript*, revised for resubmission, 2026.

Technical Expertise

Scientific ML & Deep Learning	Physics-guided representation learning; custom 1D/2D CNNs, shared-weight multi-joint encoders and late feature fusion; temporal, CWT time-frequency and FFT/spectral representations; comparative representation benchmarking; multi-seed evaluation, clustered bootstrap uncertainty quantification and robustness analysis; SVMs, PCA and t-SNE latent-space analysis.
Mathematical Foundations & Differential Geometry	Linear and tensor algebra, multivariable calculus, ordinary and partial differential equations, Fourier and Laplace methods, numerical methods, probability and statistics, calculus of variations, constrained and convex optimization; smooth manifolds, tangent and cotangent bundles, differential forms, Lie brackets, Riemannian metrics, Lie groups and Lie algebras, exponential maps, and differential-geometric modelling.
Analytical Mechanics & Energy-Based Physical Modelling	Variational, Lagrangian and Hamiltonian mechanics; symplectic and Poisson formulations, conservation laws and symmetries, rigid-body and multibody mechanics; port-Hamiltonian systems, Dirac structures, passivity, energy shaping, energy tanks, and power/energy-based modelling.
Nonlinear Systems, Control & Optimization	Linear and nonlinear dynamical systems, state-space modelling, equilibrium and stability analysis, Lyapunov stability, LaSalle invariance, controllability and accessibility, geometric control, differential flatness, optimal control, MPC, Quadratic Programming (QP), trajectory optimization, inverse dynamics, and active fault-tolerant control.
State Estimation, Signal & Image Processing	Bayesian estimation, Lie-group and invariant state estimation, SLAM foundations, Fourier and spectral analysis, digital filtering, Continuous Wavelet Transform (CWT), spatial and frequency-domain image processing, feature extraction, pattern classification, and computer-vision foundations.
Scientific Computing & Engineering Tools	Python, PyTorch, MATLAB, Simulink, Simscape Multibody, Git, $\LaTeX$, CAD/CAE (SolidWorks, Inventor, ANSYS), instrumentation, PLC/SCADA systems, and industrial communication protocols.

Academic and Professional Experience

Graduate Teaching & Laboratory Assistant <i>King Fahd University of Petroleum & Minerals (KFUPM), Saudi Arabia</i> Instructing undergraduate laboratory sessions in electronics and instrumentation, delivering hands-on guidance on circuit analysis, operational amplifiers, signal conditioning, and electronic measurement systems.	2025–Present
Teaching Assistant – Statistics and Probability <i>Escuela Superior Politécnica del Litoral (ESPOL), Ecuador</i> Assisted undergraduate engineering courses; conducted problem-solving recitations on probability distributions, parameter estimation, and statistical data analysis; evaluated student assignments.	2022
Undergraduate Research & Prototyping Assistant <i>Escuela Superior Politécnica del Litoral (ESPOL), Ecuador</i> Contributed to design and hardware prototyping for mechatronic machinery projects, assisting in CAD modeling, sensor interfacing, and PLC-based automated control routines.	2021–2022
Maintenance and Automation Coordinator <i>Detecuator S.A., Ecuador</i> Supervised automated operations, industrial instrumentation, and predictive maintenance protocols for manufacturing facilities; engineered 2D/3D CAD solutions for electro-mechanical system modernization.	2021–2024
Assistant Electromechanical Designer <i>SEOR S.A., Ecuador</i> Developed engineering blueprints, electrical control cabinets, and automation layouts for industrial machinery and packaging lines.	2023–2024

Honors and Awards

Graduate Full Scholarship , King Fahd University of Petroleum & Minerals (KFUPM)	2025–Present
Best Mechatronics Thesis Award (1st Place) , Escuela Superior Politécnica del Litoral	2022
Best University Thesis Award (2nd Place Institution-Wide) , ESPOL	2022
Academic Excellence Recognition (Top Score) , National University Admission Exam (SENESCYT)	2017

Languages

Spanish – Native	English – Advanced / Academic working proficiency
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